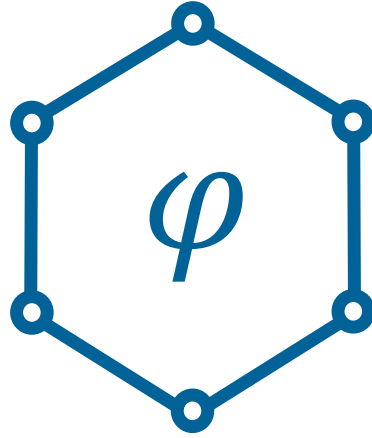
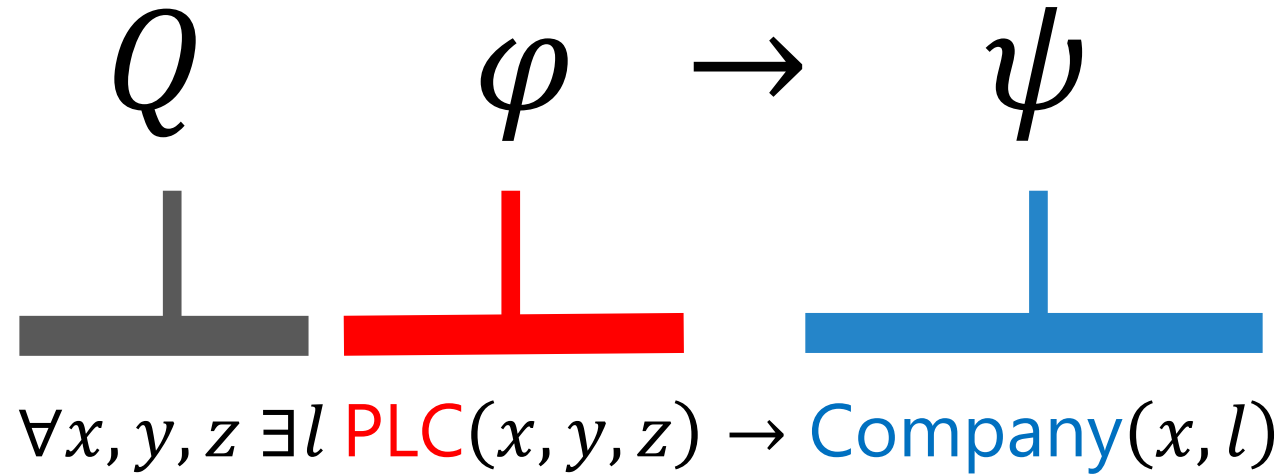




Logical Knowledge in KGs

Warded and Vadalog

Emanuel Sallinger



Knowledge

"Dependency"

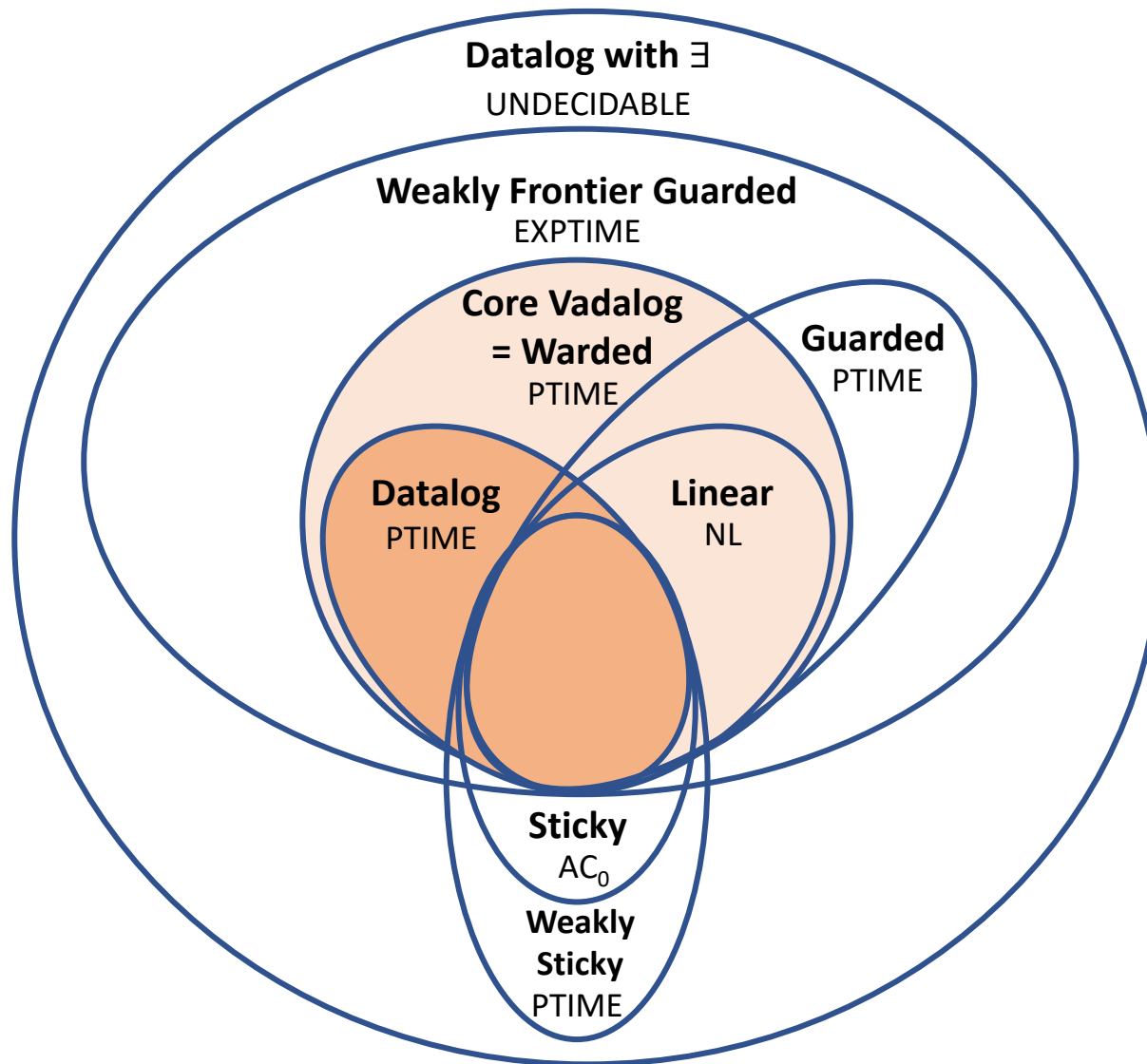
"Rule"

"Logical Formula"

"Ontology **Axiom**"

"Constraint"

"Named **Query**"



Not shown:
Shy



Language Requirements

1. **Recursive** Reasoning:
Full support of recursive **Datalog**
2. **Ontological** Reasoning:
Expressive power of **SPARQL** and **OWL 2QL**
3. **Tractable** Reasoning: **polynomial time**,
sub-fragments that are **fully parallelizable**



$$\text{Emp}(x) \rightarrow \exists z \text{ Mgr}(x, z) \quad \text{Mgr}(x, y) \wedge \text{Pers}(x) \rightarrow \text{Emp}(y)$$



$$\text{Emp}(x) \rightarrow \exists \text{z} \text{Mgr}(x, \text{z}) \quad \text{Mgr}(x, y) \wedge \text{Pers}(x) \rightarrow \text{Emp}(y)$$



$$\text{Emp}(x) \rightarrow \exists \mathbf{z} \text{ Mgr}(x, \mathbf{z}) \quad \text{Mgr}(x, \mathbf{y}) \wedge \text{Pers}(x) \rightarrow \text{Emp}(\mathbf{y})$$



Dangerous

$\text{Emp}(x) \rightarrow \exists \mathbf{z} \text{ Mgr}(x, \mathbf{z}) \quad \text{Mgr}(x, \mathbf{y}) \wedge \text{Pers}(x) \rightarrow \text{Emp}(\mathbf{y})$



$$\text{Emp}(x) \rightarrow \exists \textcolor{red}{z} \textcolor{blue}{Mgr}(x, \textcolor{red}{z}) \quad \textcolor{blue}{Mgr}(x, \textcolor{red}{y}) \wedge \textcolor{red}{Pers}(x) \rightarrow \textcolor{red}{Emp}(\textcolor{red}{y})$$

Dangerous

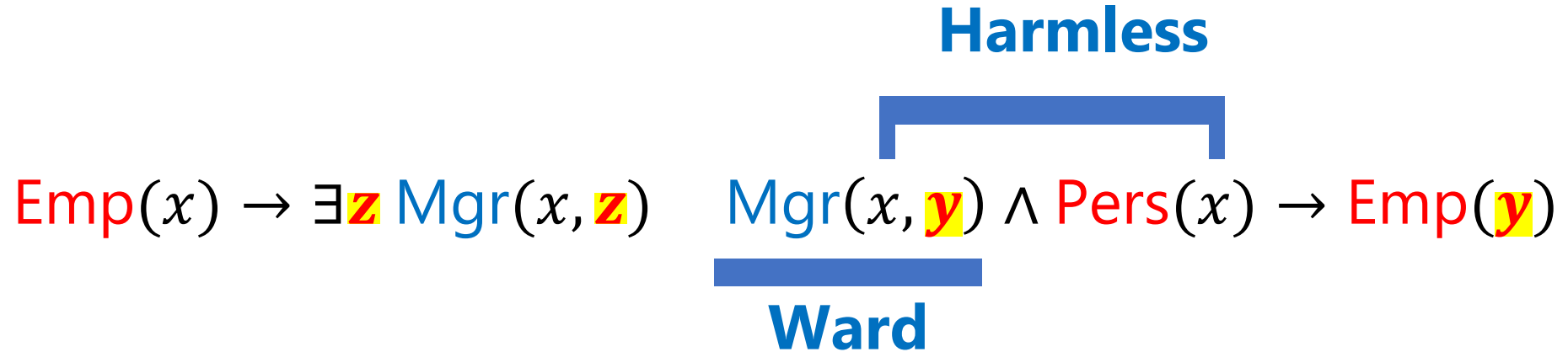
Ward

1. all the “**dangerous**” variables should coexist in a single body-atom α , called the **ward**



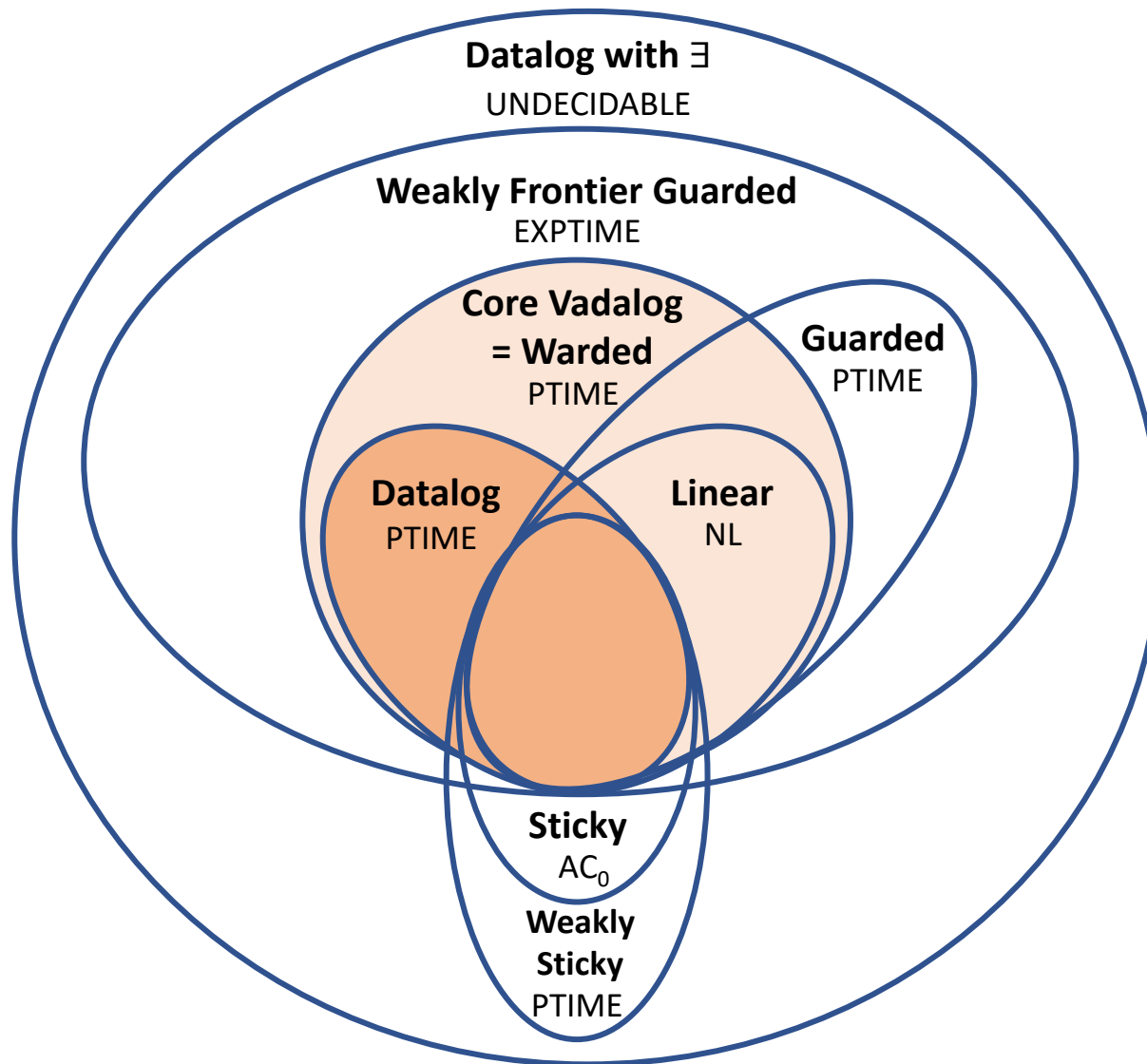
$$\text{Emp}(x) \rightarrow \exists \textcolor{red}{z} \textcolor{blue}{Mgr}(x, \textcolor{red}{z}) \quad \frac{\textcolor{blue}{Mgr}(x, \textcolor{red}{y}) \wedge \textcolor{red}{Pers}(x) \rightarrow \textcolor{red}{Emp}(\textcolor{red}{y})}{\textcolor{blue}{Ward}} \quad \textcolor{blue}{Harmless}$$

1. all the “**dangerous**” variables should coexist in a single body-atom α , called the **ward**
2. the ward can share only “**harmless**” variables with the rest of the body



1. all the “**dangerous**” variables should coexist in a single body-atom α , called the **ward**
2. the ward can share only “**harmless**” variables with the rest of the body

Warded Dependencies

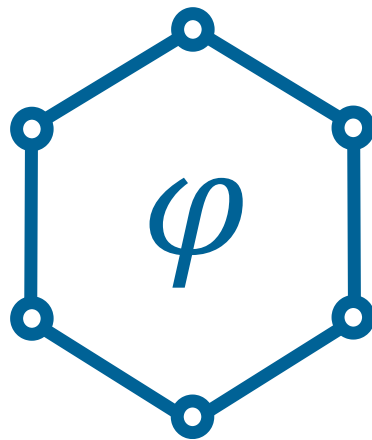


Not shown:
Shy



Main Challenges

1. **Recursion**
unlimited graph exploration
2. **Object Creation**
exploring unknown parts of the KG



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